

Non-Equivalence of Quantum Magic Squares

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0	0	0
1	0	1
0	1	?

A *nonlocal game* $G(a, b, x, y)$ is played :

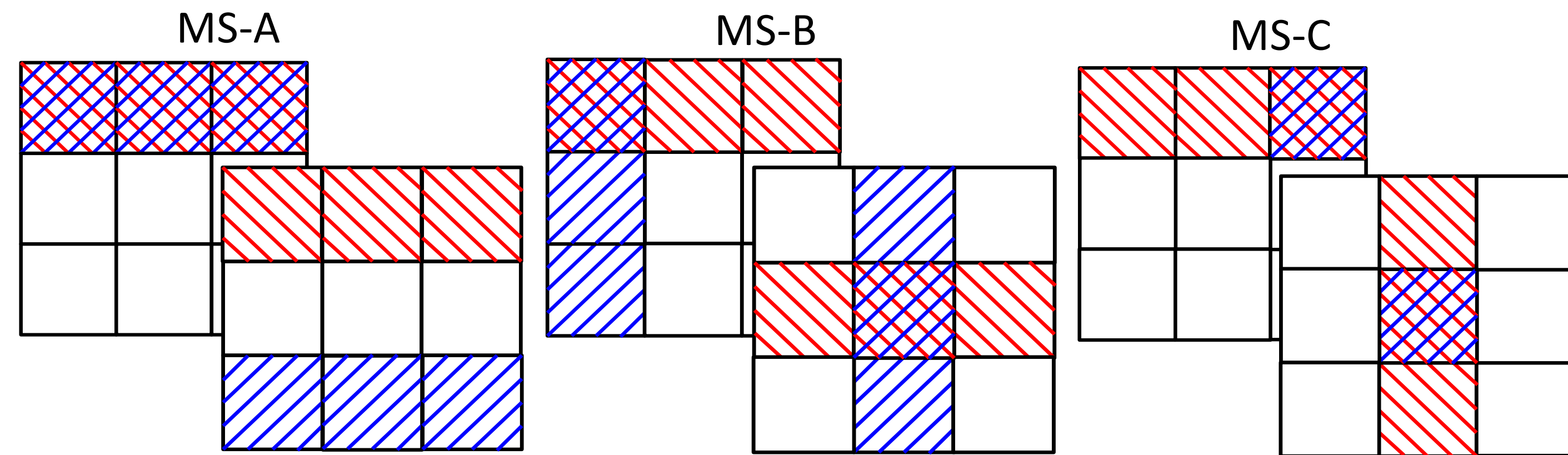
- Give input $a \in A$ to player **Alice**,
- Give input $b \in B$ to player **Bob**,
- Ask them to return $x \in X$ and $y \in Y$ respectively, without communicating

The players win the game if and only if (a, b, x, y) satisfies the predicate G .

Non-communicative strategies for players include preemptively creating a *local deterministic* strategy, sharing an entangled state (*quantum strategy*), and using a *supra-quantum* channel.

When a non-local game can be won 100% of the time using a quantum strategy, but not with a local strategy, we say that it is **pseudo-telepathic**. The only classical explanation would be telepathy between players!

A well-known pseudo-telepathic game is the **quantum magic square**. Variants of this game differ in input/output types, but all task players to fill their inputs with binary numbers such that the sum of any row is even, sum of any column is odd, and their intersections agree.



Variants of the *Quantum Magic Square* game on a 3x3 grid

Background

All QMS variants are winnable with a strategy that sees players share an entangled qubit system and, as such, are usually taken to be interchangeable.

	MS-A	MS-B	MS-C
Alice	row/column	row	row/column
Bob	row/column	column	entry in Alice's line

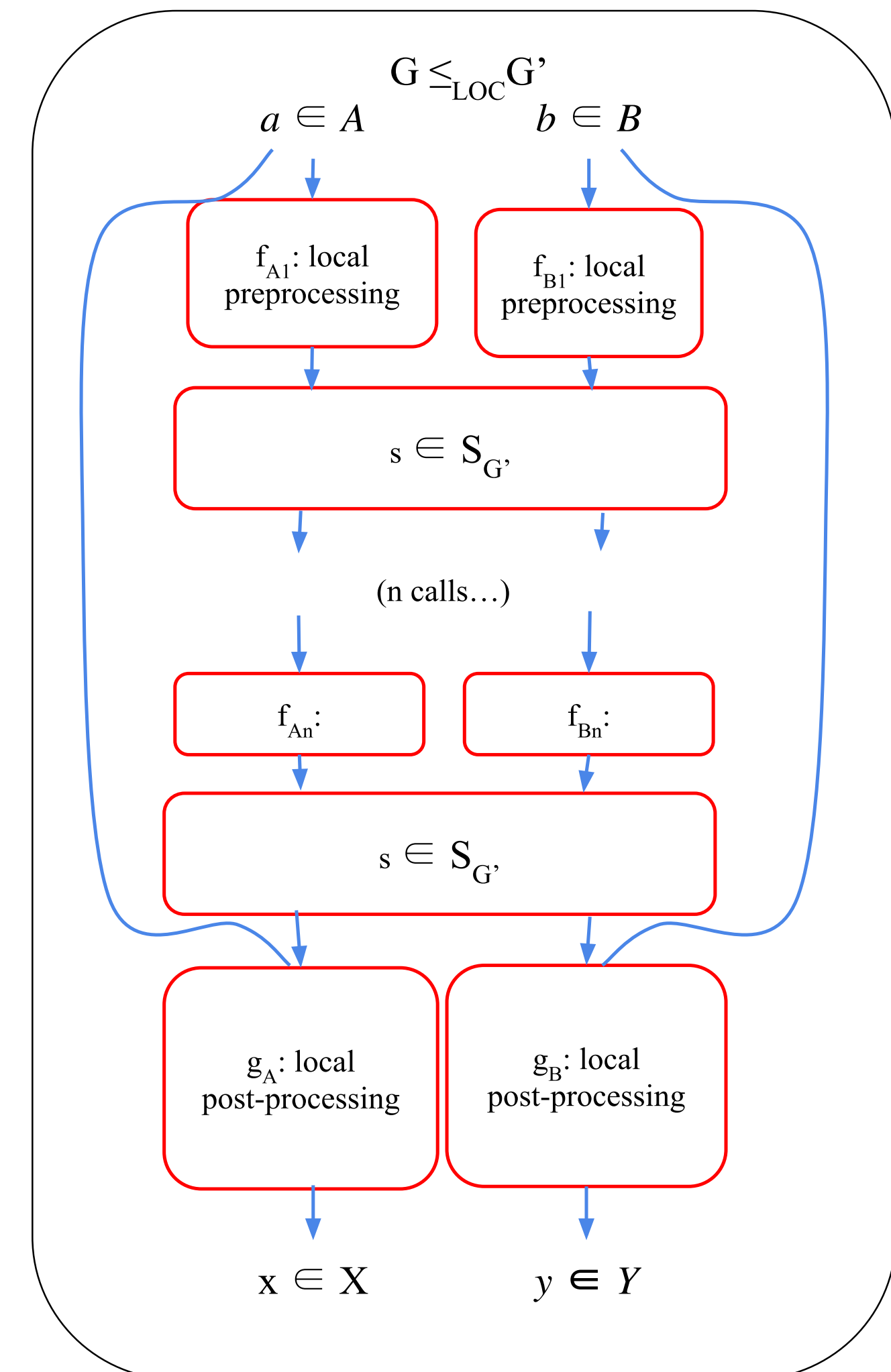
Player Input Types by Game Variant

$$[|\psi\rangle = \frac{1}{2}|0011\rangle - \frac{1}{2}|0110\rangle - \frac{1}{2}|1001\rangle + \frac{1}{2}|1100\rangle]$$

Entangled qubit system for QMS strategy

For non-local games, G and G' , we say $G \leq_{\text{LOC}} G'$ (G locally reduces to G') if we can win G perfectly using:

- An unbounded number of calls to a black box winning strategy for G'
- Local preprocessing, f , before each black box call
- Local postprocessing, g , after final black box call



A local reduction from G to G' using n calls

Goals

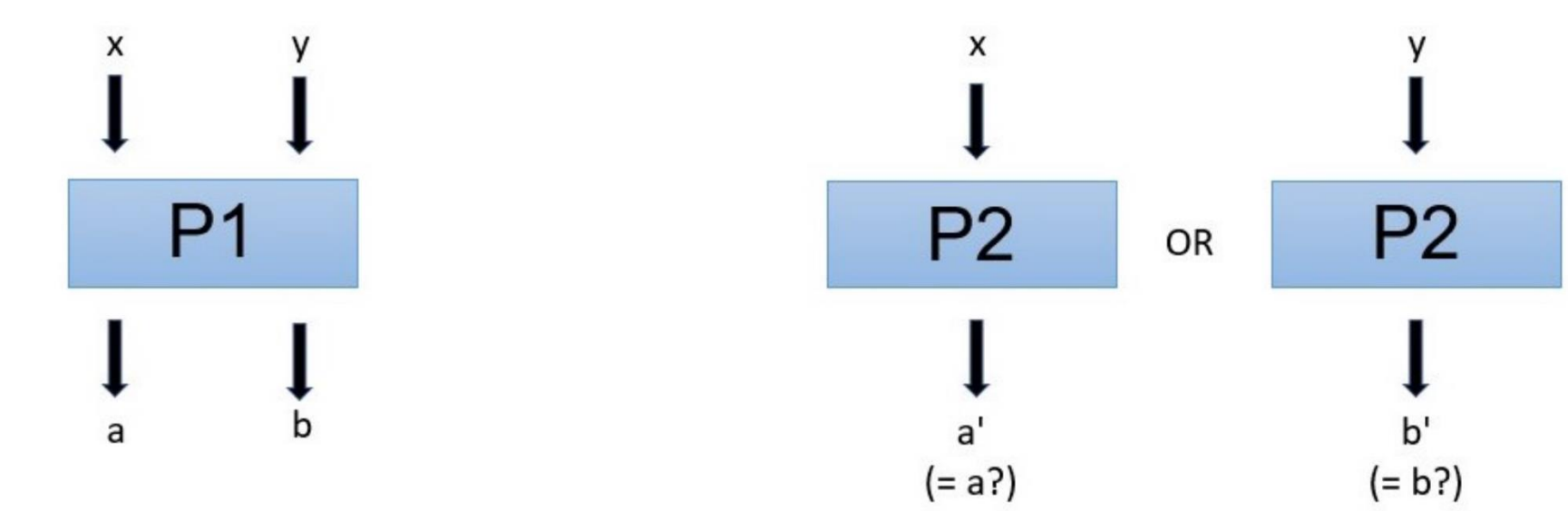
- Establish non-reductions that hinge on qualitative game properties rather than asymptotic results
- Identify pairwise reducibility relations between magic square variants
- Argue, under a reasonable definition of equivalence, that all three variants are pairwise inequivalent

We utilize the following properties of games:

- **Symmetry:** A nonlocal game G is **LOC-symmetric** if $G \leq_{\text{LOC}} \bar{G}$, where $\bar{G}$ is the *player swapped* version of G
 - Alice now receives input $b \in B$ and returns $y \in Y$
 - Bob now receives input $a \in A$ and returns $x \in X$
- **Synchronicity:** A game G is **synchronous** if, on input (a, a) , $G(a, a, x, y)$ is winning if and only if $x = y$
 - When players receive identical inputs, they must return identical outputs
- **Oracularizability:** The **oracularization** of a two-player game is its transformation to a multi-prover interactive proof system, where one player behaves as an *oracle*. A game is **oracularizable** if it simulates a proof system that is both *complete* and *sound*

Methodology

G_n^{Orac}



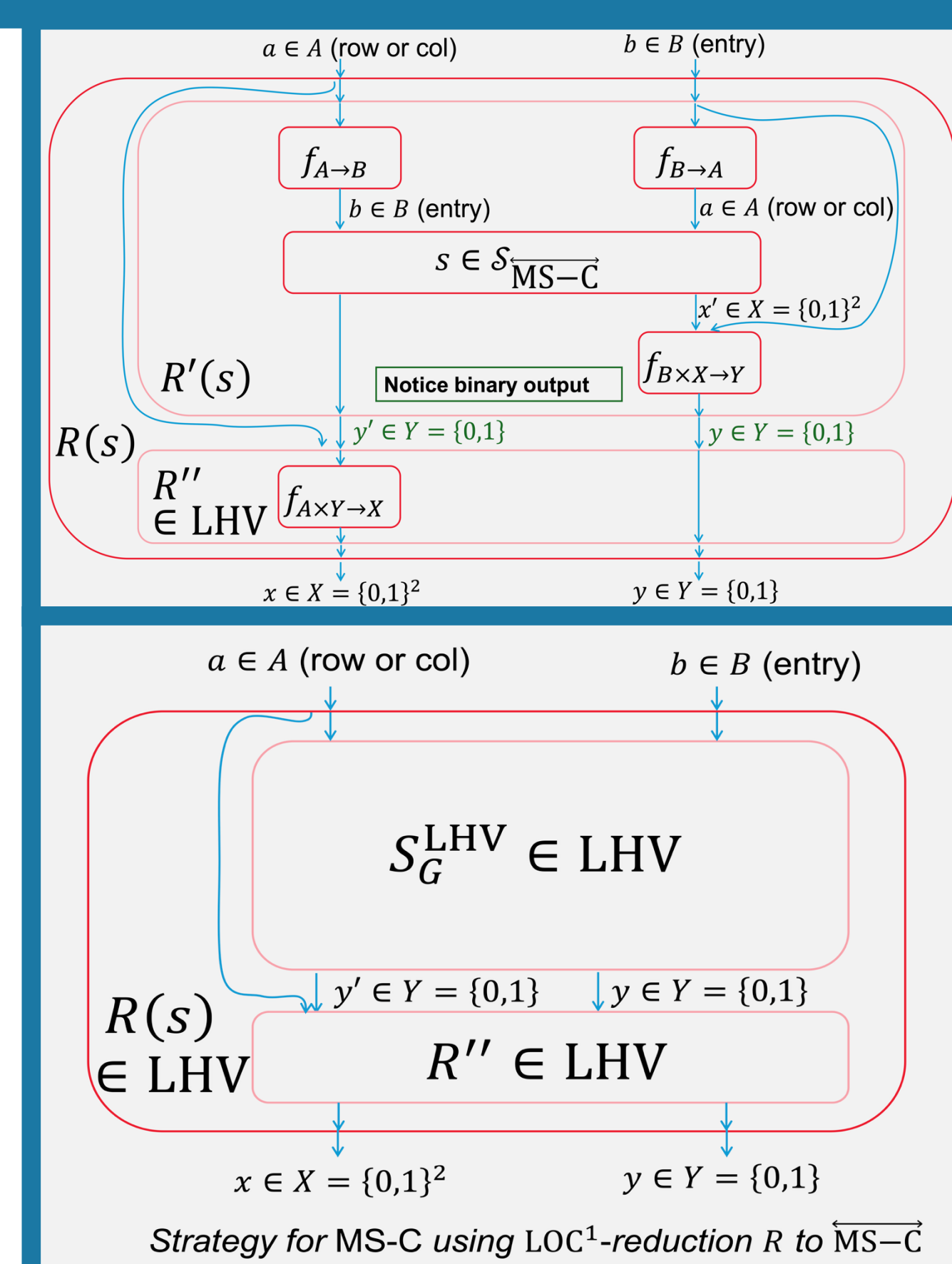
General oracularization transformation

Results

THEOREM 1: $\text{MS-A} \not\leq_{\text{LOC}} \text{MS-C}$

Proof:

- Suppose MS-C was LOC-symmetric via an unbounded number of calls to black box strategy $s \in S_{\text{MS-C}}$
- Then post-processing g_A could be included in final call to black box strategy, so last strategy has binary output for both players
- Then, last call to the black box strategy is binary, can be simulated by a local strategy
- Additionally, *sequential*, *parallel*, and *causal order* affected calls reduce to sequential calls when unbounded calls to black box are allowed
- Inductively, all previous calls to the black box strategy can be simulated by a local strategy
- Then MS-C is not LOC-symmetric while MS-A is



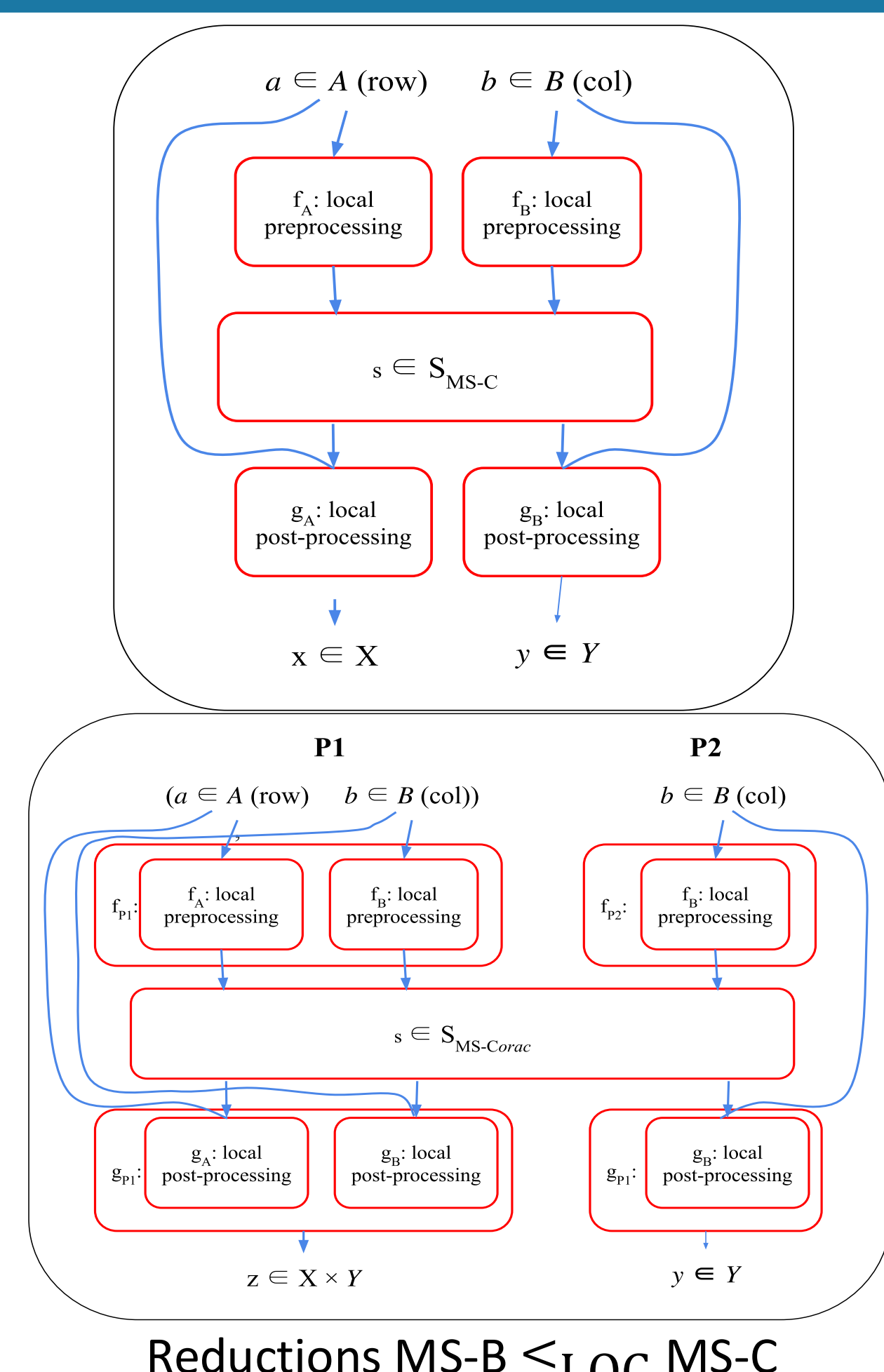
$\therefore \text{MS-A} \not\leq_{\text{LOC}} \text{MS-C}$ $\square$

THEOREM 2: $\text{MS-B} \not\leq_{\text{LOC}} \text{MS-C}$

Proof:

- Suppose $\text{MS-B} \leq_{\text{LOC}} \text{MS-C}$ via an unbounded number of calls to black box strategy $s \in S_{\text{MS-C}}$
- Then, $\exists$ sets of local processing functions, f_A, f_B, g_A, g_B , used in this reduction
- The same f_A, f_B, g_A, g_B can be reused to form a reduction $\text{MS-B}_{\text{ORAC}} \leq_{\text{LOC}} \text{MS-C}_{\text{ORAC}}$
- This is a sound reduction as $\text{MS-C}_{\text{ORAC}}$ simulates a *sound* proof system
- This presents a *complete* strategy for $\text{MS-B}_{\text{ORAC}}$
- This is a contradiction, it is known that MS-B is not oracularizable (cannot simulate a complete proof system)

$\therefore \text{MS-B} \not\leq_{\text{LOC}} \text{MS-C}$ $\square$



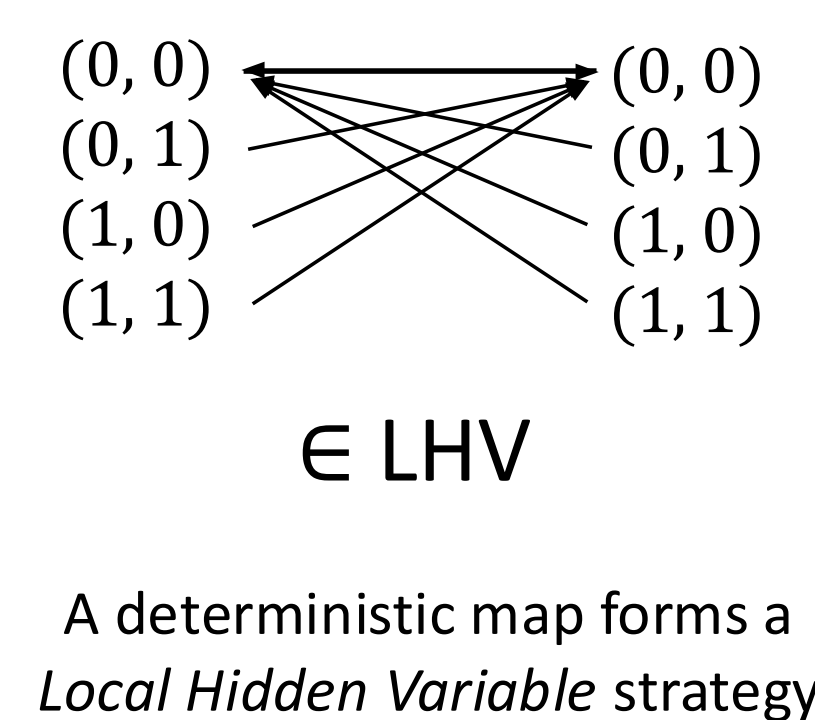
Reductions $\text{MS-B} \leq_{\text{LOC}} \text{MS-C}$

THEOREM 3: $\text{MS-A} \not\leq_{\text{LOC}} \text{MS-B}$

Proof:

- Suppose $\text{MS-A} \leq_{\text{LOC}} \text{MS-B}$ via an unbounded number of calls to black box strategy $s \in S_{\text{MS-B}}$
- On input (a, a) , post-processing g must map both *winning* and *losing* outputs of MS-B to exclusively winning
 - By **non-synchronicity**, both winning and losing outputs of MS-B are valid
- Because of finite output space, g must map all black box outputs for a given game input to a single output
- Then the reduction strategy can be represented by a deterministic function of game inputs, is local strategy
- Then MS-A is winnable by a local strategy

$\therefore \text{MS-A} \not\leq_{\text{LOC}} \text{MS-B}$ $\square$



A deterministic map forms a *Local Hidden Variable* strategy

Def: MS-AB

Let **MS-AB** be a new QMS variant with input space of MS-A and winning predicate of MS-B

- Alice and Bob both receive a row or column as input
- Input cases other than (row, col), not valid for MS-B, automatically win
- Input cases valid for MS-B must satisfy the MS-B predicate

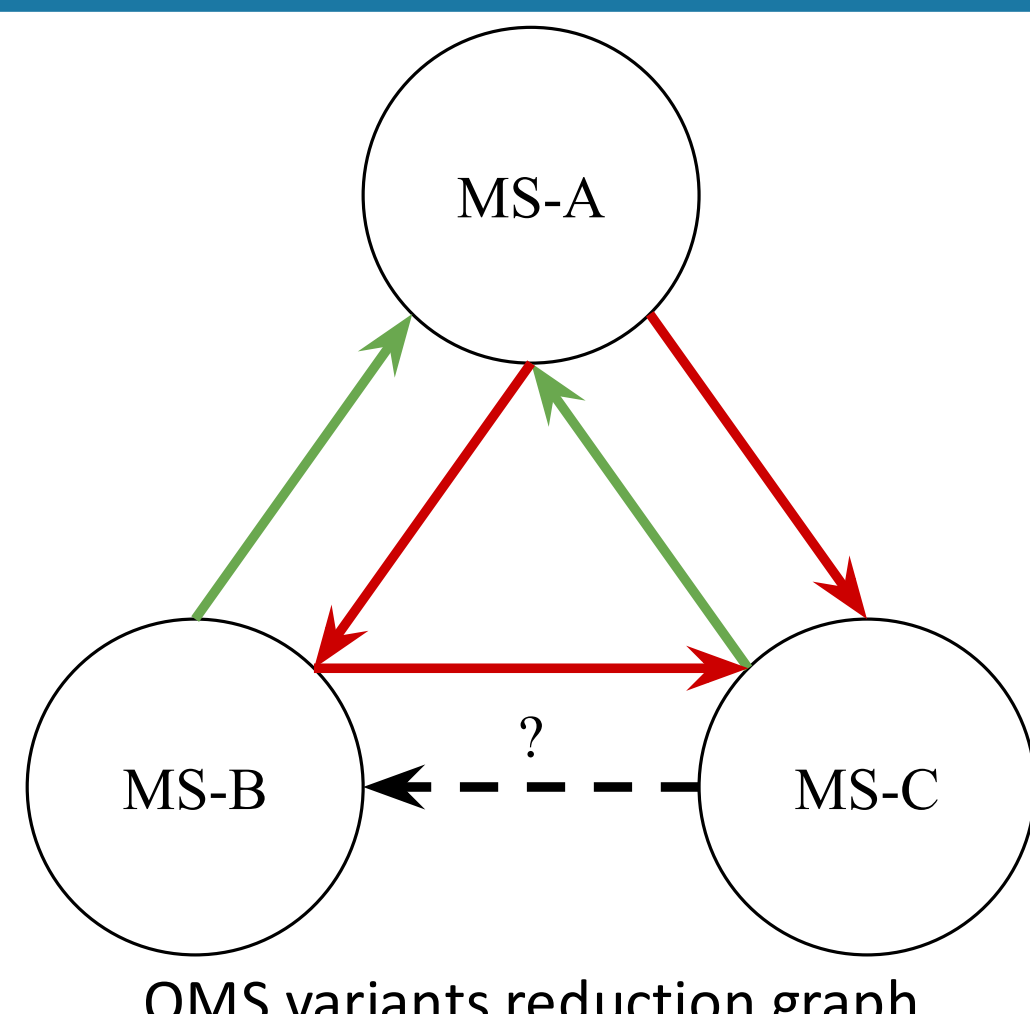
Note: $\text{MS-AB} =_{\text{LOC}} \text{MS-B}$

Conclusion

We combine results with the following previously known reduction relations, both of which are achievable via one black box call:

$\text{MS-B} \leq_{\text{LOC}} \text{MS-A}$ & $\text{MS-C} \leq_{\text{LOC}} \text{MS-A}$

For two games in the adjacent reduction graph, G and G' , a **green arc** from G to G' represents $G \leq_{\text{LOC}} G'$, while a **red arc** from G to G' represents $G \not\leq_{\text{LOC}} G'$. The figure illustrates *classical differences* between variations of the magic square problem that are provably not surmountable via quantum or nonlocal strategy.



QMS variants reduction graph

Supplementary



References

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